

Jack Ntahe ICY

Principal Design Engineer

PROFILE

Highly motivated and mature civil engineer working towards professional qualification with the Institution of Civil Engineers (ICE), with a broad base of transferable skills and an impeccable employment record. Extensive CAD and highways design experience across major UK schemes for Highways England, local authorities and leading consultancies, using AutoCAD, Civil 3D, MX Road and Keysoft packages. Adopts an analytical and logical approach to problem solving to deliver a high standard of work; a self-starter who works well on own initiative and within productive teams to meet tight deadlines.

CORE SKILLS

- AutoCAD 2D / Civil 3D / MX Road / 2D & 3D drawing and model production / Contract and setting-out drawings
- Highways design / Junction design / Horizontal & vertical alignment / Earthworks modelling and volumes
- Traffic signs & road markings (TSM, TSRGD) / KeySIGN / KeyPOSTS / KeyLINES / Autodesk Vehicle Tracking
- DMRB / MCHW / LTN 1/20 / GG 119 Road Safety Audit responses / RRRAP
- Bill of Quantities / Cost estimates / Tender documentation / Specifications and technical notes
- ProjectWise / E-Vision / Project Mapper / Microsoft Office
- Geotechnical & retaining structures / Railway and water engineering / CDM regulations

PROFESSIONAL EXPERIENCE

❖ TIFAND LIMITED — *(December 2021 – Present)*

Principal Design Engineer

- Produce 2D and 3D CAD drawings and models for civil, highways and temporary works schemes using AutoCAD and Civil 3D, in line with TIFAND drawing standards.
- Prepare general arrangement, setting-out, detail and contract drawings for tender and construction issue.
- Develop horizontal and vertical alignments, earthworks models and volume calculations to support the design engineers.
- Prepare traffic signs, road markings and swept-path analysis drawings using KeySIGN, KeyLINES and Autodesk Vehicle Tracking.
- Apply drawing management and document control procedures, including revision control, title blocks and drawing registers within the Common Data Environment.

- Check drawings for accuracy, consistency and compliance with client standards, DMRB and project specifications before issue.
- Coordinate CAD information with design engineers, the CAD Manager and other disciplines, incorporating mark-ups and design changes promptly.
- Support quantity take-off and Bill of Quantities preparation from CAD models.
- Contribute to CAD standards, templates and continuous improvement of drawing production workflows.

❖ KIRKLEES COUNCIL — (June 2021 – November 2021)

Senior Highways Design Engineer

Huddersfield Southern Corridors: Longroyd & Lockwood Bar – WYCA funded (Budget £14M)

- Designed new junctions and improved existing junctions, a new segregated cycle footway and a new link road.
- Designed all traffic signs and road markings and delivered all 2D and 3D drawings.
- Prepared the BoQ, including earthworks volume calculations.
- Delivered contract drawings and specification for tender, and setting-out drawings and information for tender and construction.
- Liaised with statutory undertakers.

Transforming City Fund (TCF): Aldams Road and Bradley to Brighouse

- Prepared and issued all detailed design drawings to LTN 1/20 and contract documents to the MCHW.
- Issued the scheme 3D design, earthworks and setting out; responded to the RSA to GG 119.
- Designed road signs and road markings to TSM and TSRGD, and traffic sign posts.

Software used: AutoCAD, Civil 3D, KeySIGN, KeyPOSTS, KeyLINES and Autodesk Vehicle Tracking.

❖ ATKINS – SNC-LAVALIN — (October 2018 – June 2021)

Highways Engineer

Designer – Calderdale A646 and A58 Corridor Improvement Programme

- Prepared the scheme bid cost estimate.
- Delivered the preliminary and detailed design for all highways improvement works using AutoCAD, Civil 3D, swept path analysis and PDS signs and road markings.
- Prepared the Road Safety Audit (RSA), TROs and the Bill of Quantities for construction.
- Led and delivered the Statutory Undertakers works.

Design Lead – ELOR East Leeds Orbital Route

- Led the Atkins GDC in India to deliver preliminary and detailed designs for S400 (RRRAP and VRS) and S1100 (Footway and Cycle Superhighway).
- Agreed design timescales with internal and external bodies and reviewed, checked and coordinated design requirements with the India design hub.
- Attended meetings with clients and contractors; prepared specifications, appendices, technical notes and design input statements; responded to RSA to GG 119.

M6 Junction 20

- Prepared the highways improvement cost estimate.

❖ **JACOBS ENGINEERING — (October 2015 – October 2018)**

Civil Highways Engineer

Manchester North-West Quadrant – Highways England PCF Stage 1

- Identified potential route options for increasing capacity and resilience; data collection, analysis and technical notes.
- Produced horizontal and vertical Civil 3D designs for proposed bypass tunnel alignment options and identified M60 tie-in points.

Oxford to Cambridge (OxCam) Expressway PCF Stage 1 (Total budget £5.7 billion)

- Produced existing road geometry schematic drawings and the schedule of A34 junctions.
- Designed horizontal and vertical alignments of the A34 and OxCam interchange junction and OxCam mainline options.

Operations Directorate East Capital Delivery, Areas 6 & 8 (Budget £21M)

- Prepared design deliverables and construction documents for the A11, A12(S), A14 and A120 trunk road renewal schemes in Cambridgeshire, Essex and Suffolk.
- Prepared TTRO applications, road space bookings, environmental scoping documents, traffic management drawings and technical notes.
- Liaised with Local Authorities and Highways England to resolve work programmes and obtain authorisation to proceed.

M6 J16–19 Smart Motorway Project – Highways England (Total budget £193M)

- Assisted in re-modelling the Civil 3D scheme design into a full MX road alignment design, including earthworks and retaining walls, MCS and Emergency Refuge Areas.
- Checked pavement drawings and produced isopachyte drawings for construction.
- Reviewed the CDM Design Hazard Log and RRRAP and updated the Implementation Report for New Standards.

❖ **KELLER UK / PHI GROUP RETAINING STRUCTURES — (March 2013 – October 2015)**

Regional Geotechnical Engineer

- Delivered client requirements from initial contact, cost estimation and design through to construction.
- Ensured safe working practice of site operations and liaised with the management team.
- Prepared method statements, risk assessments, detailed AutoCAD drawings and design calculations.
- Procured materials and selected appropriate plant and equipment.
- Prepared monthly Applications for Payment, managed subcontractors and certified their invoices.
- Implemented procedures to reduce or remove health and safety hazards under CDM regulations.
- Developed client relationships to encourage repeat business and mentored less experienced colleagues.

❖ **NENE VALLEY RAILWAY (NVR) — (September 2012 – March 2013)**

Graduate Civil Engineer (Railway)

- Worked on rail engineering design for a 10-year infrastructure forward plan.
- Carried out geometry assessments of 9 miles of track.
- Designed and improved two foot/cycle level crossings and chicane barriers for a new railway line.

- Designed a reinforced concrete inspection pit and advanced designs following local authority comments.
- Gained site experience in track alignment, maintenance and levels, and assisted on lineside and S&T equipment renewals.

❖ **WERM ENGINEERING LTD – ROMFORD — (*June 2012 – September 2012*)**

Graduate Engineer (Water)

- Applied engineering solutions to practical problems and sewer flooding investigations within water and wastewater management.

❖ **CITY OF BRADFORD METROPOLITAN DISTRICT COUNCIL — (*June 2011 – September 2011*)**

Summer Placement – Major Highways Unit

- Assisted in the preliminary geometric and engineering design of the £7.5M Low Moor Rail Station and car park scheme.
- Drafted inputs for technical reports and worked with AutoCAD 2D, Bentley MX and DMRB.

EDUCATION & TRAINING

- BEng (Hons) Civil & Structural Engineering – University of Bradford (2009 – 2012)
- Certificate, Engineering Foundation Year – University of Bradford (2008 – 2009)
- HND equivalent, I.T. and Management – ISIG (1996 – 1999)
- CSCS Card / Highways England Passport Card

PROFESSIONAL MEMBERSHIPS

- Institution of Civil Engineers (ICE) – working towards professional qualification